CS 82A - Module 1 Lab 2: Textbook Problem 1.1

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Goal: compare the minimum, mean, and maximum six-week chicken weight for every feed type and use those results to make a recommendation.

1. Load the course dataset

Code

```
from pathlib import Path
import pandas as pd

data_path = Path("data/chickwts.csv")
chicks = pd.read_csv(data_path)
print(f"Rows: {len(chicks)}")
print(f"Feed types: {chicks['feed'].nunique()}")
chicks.head()
```

Output

```
Rows: 71
Feed types: 6

weight       feed
0     179  horsebean
1     160  horsebean
2     136  horsebean
3     227  horsebean
4     217  horsebean
```

2. Check the data

Code

```
assert len(chicks) == 71
assert set(chicks.columns) == {"weight", "feed"}
assert chicks.isna().sum().sum() == 0
print("Checks passed: 71 rows, expected columns, and no missing values.")
```

Output

```
Checks passed: 71 rows, expected columns, and no missing values.
```

3. Summary table

Code

```
feed_summary = (
    chicks.groupby("feed")["weight"]
    .agg(Minimum="min", Mean="mean", Maximum="max")
    .round(2)
    .sort_values("Mean", ascending=False)
)
feed_summary
```

Output

feed	Minimum	Mean	Maximum
sunflower	226	328.92	423
casein	216	323.58	404
meatmeal	153	276.91	380
soybean	158	246.43	329
linseed	141	218.75	309
horsebean	108	160.20	227

4. Evidence notes for your decision

The executed table above supports the following comparison.

- Highest mean and maximum: sunflower has a mean of 328.92 g, a maximum of 423 g, and a minimum of 226 g.

- Strong alternative: casein has a mean of 323.58 g, a maximum of 404 g, and a minimum of 216 g.

- Main comparison: sunflower and casein are close on average, so decide how much the 5.34 g mean difference and the observed minimums matter to your business priority.

- Lower-performing comparison: horsebean has the lowest mean (160.20 g) and the lowest minimum (108 g).

These are calculation notes, not a recommendation.

5. Recommendation

I recommend sunflower feed because it produced the strongest overall weight results in this sample. Its mean weight was 328.92 g, the highest among all six feeds. It also had the highest minimum weight at 226 g and the highest maximum weight at 423 g, showing strong performance across the observed chicks. Casein was the closest alternative with a mean of 323.58 g, a minimum of 216 g, and a maximum of 404 g, but it trailed sunflower on all three measures. For a business that prioritizes heavier six-week chickens while also limiting low-weight outcomes, sunflower is the better choice. These results come from this dataset, so cost and other practical factors should also be considered before making a final business decision.